

Exam #2

Math 742 - Mathematical Statistics

3/25/2026

Name_____

Directions: The in-class portion of exam 1 is worth 50 points. The exam is split into 2 sections. Part I (problems 1-3) is 50 minutes. Part II (problems 4-5) is 25 minutes. You may have a hand held calculator for the arithmetic and a 8 1/2 x 11" sheet (one-sided) of notes.

Part I - Theory + asymptotics

Problem 1 (10 points)

Let $X_1, X_2, \dots, X_n$ be an iid sample from an normal distribution with mean μ and variance $\sigma^2 < \infty$. Consider the estimator,

$$\bar{T} = 2\bar{X}$$

- a) (3 points) Find the bias of T as an estimate of μ .
- b) (4 points) Find $\text{Var}(T)$.
- c) (3 points) Compute the mean squared error of T and determine whether T is consistent for μ

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2. Let $X_1, X_2, \dots, X_n$ be an iid sample from $N(\mu, \sigma^2)$, where $\sigma^2 < \infty$ is known.
- a) (3 points) Find the Fisher information ($I_n(\mu)$) for this sample.
 - b) (3 points) State the Cramér–Rao lower bound for unbiased estimators of μ .
 - c) (4 points) Is $\bar{X}$ efficient? Justify your answer using variance comparisons and interpret what efficiency means in this setting.

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3. Let $X_1, X_2, \dots, X_n$ be an iid sample with mean μ and variance $\sigma^2 < \infty$.

Define:

$$T_n = \frac{\sqrt{n}(\bar{X} - \mu)}{S_n}$$

Where S_n^2 is the sample variance.

- a) (3 points) State the Central Limit Theorem by specifying the limiting distribution of the properly standardized sample mean $\bar{X}$.
- b) (3 points) Explain what it means for an estimator to be consistent. Then explain why S_n^2 is a consistent estimator of σ^2 .
- c) (4 points) Use Slutsky's Theorem to determine the limiting distribution of T_n .

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Part II — Hypothesis Testing

Problem 4

Suppose $X_1, X_2, \dots, X_n \sim N(\mu, \sigma^2 = 4)$

We want to test:

$$H_0 : \mu = 0 \text{ vs } H_1 : \mu > 0$$

- a) (3 points) Construct an appropriate **standardized test statistic**.
- b) (3 points) Give the rejection region for a level $\alpha = 0.05$ test.
- c) (4 points) Explain how a p-value is computed and interpret what a small p-value means in this context.

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Problem 5

Let $X_1, X_2, \dots, X_n \sim \text{Bernoulli}(p)$. Consider the following hypothesis test:

$$H_0 : p = 0.3 \text{ vs. } H_1 : p = 0.6$$

- a) (2 points) Write out the likelihood ratio:

$$\Lambda(x) = \frac{L(0.3)}{L(0.6)}$$

- b) (2 points) Using the Neyman–Pearson Lemma, explain why the most powerful test rejects for large values of $\sum_{i=1}^n X_i$.
- c) (3 points) Suppose the rejection rule is:

$$\sum_{i=1}^n X_i \geq c$$

Explain conceptually how power depends on the choice of c .

- d) (3 points) Suppose the rejection region is $\sum_{i=1}^n X_i \geq c_n$ where,

$$\frac{c_n}{n} \rightarrow k \quad \text{for some } 0.3 < k < 0.6$$

Is this sequence of tests consistent against the alternative $p = 0.6$? In other words, does the power converge to 1 as $n \rightarrow \infty$?

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